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Device Simulation

This is a two-dimensional cross section of a fluid carrying pipe. The fluid, whether it be oil, brine, or some mixture thereof, occupies the top region. The bottom region features an insulator which is abutted on three sides by a single conductor. In addition, there is a transmitting electrode surrounded by the insulator on three of its four faces, with the final side bounded by the fluid.

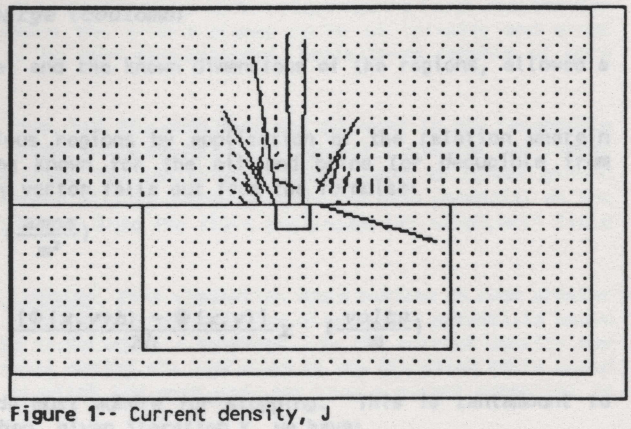


Figure 1-- Current density, J

For this particular simulation, I opted for the transmitting electrode to be at two volts and the surrounding conductor to be a ground plane. The insulator is Teflon, with a dielectric constant of 2.1 and a resistivity of some 10^{7} ohm-m. The fluid here is a brine/oil mixture with a composite dielectric constant of 30 and a resistivity of 1,000 ohm-m.

In arriving at these results, I opted for an electrostatic treatment of the problem. This means that I endeavored to solve Laplace's equation globally; the equation is namely:

$$\nabla^2\Phi = 0$$

$$\Phi = \Phi(x,y) \doteq potential(volts)$$

The graph describes the current density induced by this arrangement. I discretized the above problem by use of a Taylor expansion about each of my intended grid points. For a small h, I found that:

$$\Phi(x+h,y) = \Phi(x,y) + h\frac{\partial\Phi(x,y)}{\partial x} + h^2\frac{\partial^2\Phi(x,y)}{\partial x^2} + \ldots + h^n\frac{\partial^n\Phi(x,y)}{\partial x^n}$$

This equation was subsequently trunctated such that only terms to the second order were retained. This means that my expansion about (x+h) entailed:

$$\Phi(x+h,y) \approx \Phi(x,y) + h\frac{\partial\Phi(x,y)}{\partial x} + h^2\frac{\partial^2\Phi(x,y)}{\partial x^2}$$

In addition, I carried out this partial expansion for the potential at (x-h), (y+h), and (y-h). With these in hand, I collected terms and found the ensuing approximation to Laplace's equation:

$$\nabla^2\Phi = \frac{\partial^2\Phi}{\partial x^2} + \frac{\partial^2\Phi}{\partial y^2} = \frac{\Phi(x,y+h) + \Phi(x,y-h)}{2h^2}$$

The above formulation is fine for a homogeneous region within which to carry out the calculation, but the dielectric step from the insulator to the fluid

$$+ \frac{\Phi(x-h,y) + \Phi(x+h,y) - 4\Phi(x,y)}{2h^2}$$

necessitated a formulation with faster convergence characteristics. In order to satisfy this need, I employed an accelerating factor. With this factor, Ω, I could augment my rate of convergence by several factors of computation time. Ω is a real number between one and two, with one being the case of simple averaging. This is a technique which falls under the rubric of Successive Over Relaxation. The altered version of Laplace's Equation appears as follows:

$$\Phi^{(k+1)}(x,y) = \Phi^{(k)}(x,y) + \frac{\Omega}{4}\left(\frac{h^2\rho(x,y)}{\epsilon_r\epsilon_o} - 4\Phi^{(k)}(x,y)\right.$$

$$\left. + \Phi^{(k)}(x+h,y) + \Phi^{(k)}(x-h,y) + \Phi^{(k)}(x,y-h) + \Phi^{(k)}(x,y+h)\right)$$

This is actually Poisson's equation, as it entails the space charge density ρ(x,y) (coulomb/m^3). This density assumes its maxima about the edges of the conducting surfaces, where opposite polarities induce charge flow.

Poisson's equation is listed below.

$$\nabla^2\Phi = -\frac{\rho}{\epsilon_r \epsilon_o}$$

In order to find the charge on the conductors, I discretized Gauss' Law by way of integrating over the charged surfaces in order to find Q, the charge therein:

$$Q = \sum_x \sum_y \frac{\partial \Phi}{\partial n} = charge\ (coulomb)$$

where n is the unit normal to the surface. This charge, and the known dimensions of the regions, allowed a formulation of $\rho(x,y)$ for each computation.

In addition, I found the current density in the various regions by application of the relation wherein conductivity σ = 1/resistivity, with the latter being known for the oil and brine (or deducible from measurements to follow). From this, the current density vector falls out from the formulae:

$$\mathbf{J} = \sigma_c \mathbf{E} \quad \left(\frac{amps}{m^2}\right)$$

where

$$\mathbf{E} \triangleq -\nabla\Phi = \frac{[\Phi(x+h,y) - \Phi(x,y)]}{2h}\mathbf{x} + \frac{[\Phi(x,y+h) - \Phi(x,y)]}{2h}\mathbf{y} \quad \left(\frac{volts}{m}\right)$$

From here, I specified the tolerance to which the code must adhere for stopping. This is tantamount to prescribing a small number δ such that the code stops when, given iteration k, we have:

$$\forall x \forall y \quad |\Phi^{(k+1)}(x,y) - \Phi^{(k)}(x,y)| < \delta$$

The tolerance δ should be chosen to be no less than h^2 as the area of each mesh in the simulation bounds the error in some sense. The simulation will only cease once each resulting mesh potential and its former differ by the prescribed tolerance. This can lead to a large number of iterations as the code allows up to ten thousand mesh points.

Regarding output, the user has manifold options. Upon completing a successful simulation, a screen picture of the mesh resolution can be obtained. Then, a picture of the actual geometry with either the scaled current density vectors J or the electric field strengths E superimposed will be offered.

As a simple picture cannot often suffice to provide exact answers, the data regarding the fields and potentials at each grid point will be sent to separate text files. Because of their size and format, it is necessary to arrange to data with a program such as Lotus 1-2-3. This data can subsequently be plotted with a package such as SigmaPlot in order to yield level curves relating to the field strengths or potentials at various radii in the pipe.

In addition to this, elementary considerations are employed to find the impedance and capacitance of the simulation. I first modelled the configuration a being parallel plate capacitors, with the insulator and fluid acting as the resident dielectric of each. These were then in turn treated as being in parallel, taking the respective geometry into consideration.

This is fine, but the model falls apart as the transmitting electrode and bounding conductor are not of negligible thickness. To ameliorate this situation, I assumed that the capacitance C can be culled from the following formulation:

$$C = Q\,\underline{\Phi}_t^{-1} \quad (Farads)$$

where $\underline{\Phi}_t$ is the potential difference between the conductors (which has been formulated as a 2-dimensional matrix per our problem). This then leads to the impedance Z_o from the relation:

$$Z_o = \frac{1}{v_o\sqrt{CC_o}} \quad (ohms)$$

where C is the capacitance in the absence of the dielectrics and C_o is the capacitance with the dielectrics present. v is the phase velocity in the simulation, where v can be found from:

$$v = v_o\sqrt{\frac{C_o}{C}} = \frac{c}{\sqrt{\epsilon_r}} = phase\ velocity\ \left(\frac{m}{s}\right)$$

We have c as the speed of light (3×10^8 m/s) and ϵ_r as the dielectric constant. This naive formulation does not quite hold physically. In fact, a pure TEM wave cannot be supported in a geometry where there exists discontinuous media over the cross section. But this fact, with its attendant frequency dependence, ought not

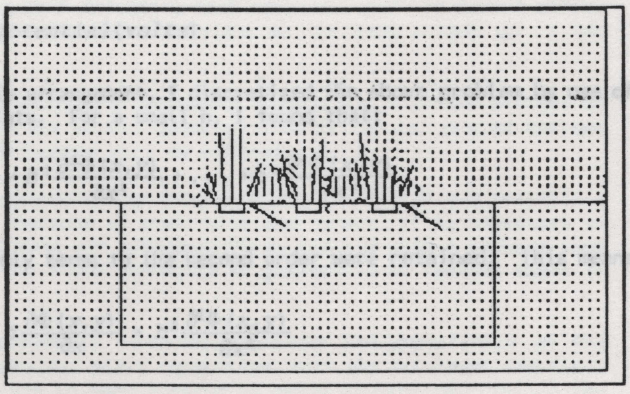
to create too much error in the simulation. Any such error should be small with respect to discretization error, which has been noted to be of $O(h^2)$.

Although this simulation only features two conductors, it can be a first approximation to the spherical focusing arrangement, which is needed to invade a highly resistive fluid (e.g., oil). To do so, the two conductors are specified to be at the same potential, therein giving the current from the center electrode no local destination but only a path deeper into the fluid. In fact, assuming the ground planes to be at infinity and minus infinity, the field lines from the center electrode will be more concentrated, hence being capable of more thorough invasion.

Sadly, this assumption of a ground plane at infinity is fine, but the code does not converge in any reasonable time. The only way to get the fast convergence of the former simulation is to explicitly specify additional transmitting electrodes which will be separate from but flank the center electrode. Then, my former ground plane can retain its potential at zero, and the new electrodes will be energized with the same potential as the center one, thus giving the center one's current no path but up into the fluid and around its neighbors' field lines.

Per the discussion in Jay Tittman's text Geophysical Well Logging, this simulation would achieve an even greater depth of investigation by maintaining a constant power. That is, the transmitting electrodes' potentials would become variables so that current multiplied by potentials would equal a constant. As the current density was a measured quantity, this would lead to varying the potential during the evolution of one iteration (this would only be necessary for a time-varying media, or a media to which the user does not know a proper potential to specify on his transmitting electrodes).

Regarding the use of guard electrodes to focus the current into the formation, two additional electrodes have been added to the simulation. These two conductors can be specified to maintain a certain potential. Their location and dimensions are also at the liberty of the user. A typical simulation with the three electrode arrangement is pictured in Figure-2.



This simulation was programmed in Turbo Pascal and has about 1800 lines (34 page printout). It has been written such that the user can specify the insulator, conductor, and their various physical parameters (e.g., dielectric constant, resistivity, skin depth, etc.). Also, there are many options for the fluid under test. It can be oil or brine with various parameters, plus the two can be combined into either water or oil continuous flow. In addition, simple mixing relations and temperature dependence are postulated so as to give the user to most freedom in configuring the tool.

Beyond the materials, the user must also specify every dimension of the tool and provide the following numerical parameters: step size, relaxation factor and stopping tolerance, all of which best suit his or her needs.

Last of all, there is an option to specify default values for all of the simulation parameters if it becomes tiresome to continually program the same scenario with only slight modifications from one sitting to the next.